

Quantum Algorithm for Computing the Period Lattice of an Infrastructure

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Joint work

- *Quantum Algorithms for One-Dimensional Infrastructures*

with Pradeep Sarvepalli (University of British Columbia)
preprint arXiv:1106.6347

- *Quantum Algorithm for Computing the Period Lattice of an Infrastructure*

with Felix Fontein (University of Zürich)
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Motivation and overview

- infrastructures are group-like algebraic objects
- they appear in arithmetic geometry in the study of computational problems related to algebraic number fields and function fields with finite constant fields
- arithmetic geometry provides a unified treatment of global fields
- the most prominent computational tasks are computing
 - the period lattice of the infrastructure and
 - generalized discrete logarithms
- both problems are not known to have efficient classical algorithms

QA for problems in algebraic geometry

- there is an elegant and efficient quantum algorithm for infrastructures arising from function fields (of curves over finite fields)
- such infrastructures are discrete and embed into finite abelian groups (divisor class groups of degree zero)
- there are efficient (classical) methods to do arithmetics in these groups
- $\Rightarrow$ we can simply use the quantum algorithm for the abelian HSP to compute the period lattice of discrete infrastructures

QA for problems in algebraic geometry

- the quantum algorithm for the abelian HSP can also be used to efficiently
 - compute generalized DLOGs in discrete infrastructures
 - compute the whole divisor class group and the ideal class group
 - solve the principal ideal problem
 - compute the Zeta function
- $\Rightarrow$ we focus our attention on non-discrete infrastructures
- such infrastructures arise from number fields

Abstract infrastructures

- an n -dimensional infrastructure $\mathcal{I}$ is a finite set of distinguished points on an n -dimensional torus $\mathbb{R}^n / \Lambda$
- Λ is a lattice of full rank in $\mathbb{R}^n$
- to each distinguished point x , we assign a region on the torus, so that every point on the torus lies in exactly one such region
- $\Rightarrow$ every point y in the region of x can be represented by the difference $t := y - x$ together with x , i.e. as the pair (x, t)
- these tuples (x, t) are called the f -representations of the infrastructure

Group arithmetic with f -reps.

- denote the set of all f -representations by $\text{Rep}(\mathcal{I})$
- $\text{Rep}(\mathcal{I})$ is a subset of $X \times \mathbb{R}^n$
- X is some finite set and its elements label the distinguished points of the torus
- $\text{Rep}(\mathcal{I})$ can be endowed with a group structure such that

$$\text{Rep}(\mathcal{I}) \cong \mathbb{R}^n / \Lambda$$

HSP over $\mathbb{R}^n$

- there is a surjective group homomorphism

$$f : \mathbb{R}^n \rightarrow \text{Rep}(\mathcal{I}) \quad \text{with} \quad \ker f = \Lambda$$

- $\Rightarrow$ intuition: determine the subgroup Λ of $\mathbb{R}^n$ hidden by f
- but there are two fundamental problems:
 - we can only evaluate at a finite number of points
 - we cannot compute f exactly because we cannot
can only approximate f -representations
- $\Rightarrow$ we have to work with a finite number of carefully
chosen evaluation points that avoid “forbidden regions”

Forbidden regions

Approximate homomorphism f

- let $\mathcal{V} = \{0, \dots, qN - 1\}^n$
- let S be a finite subset of $X \times \mathbb{N}$ related to $\text{Rep}(\mathcal{I})$
- $\Rightarrow$ we can efficiently compute a function $f : \mathcal{V} \rightarrow S$ with

$$f(v) = f(v') \Leftrightarrow \frac{v - v'}{N} \approx \lambda \text{ for some } \lambda \in \Lambda$$

- there is a collision iff v and v' differ approximately by an integer multiple of a lattice vector of the period lattice Λ
- the elements of the preimage $f^{-1}(v) \subset \mathcal{V}$ have the special form

$$v' = v + N\lambda + \xi_\lambda \quad \text{with} \quad \xi_\lambda \in (-1, 1)^n$$

Pseudo-periodic states

- let Λ be a full-rank lattice in $\mathbb{R}^n$
- let $\mathcal{V} = \{0, \dots, qN - 1\}^n$
- a uniform superposition in $\mathbb{C}^{\mathcal{V}} := (\mathbb{C}^{qN})^{\otimes n}$ over a set of the form

$$\mathcal{M} = \{v' = v + N\lambda + \xi_\lambda \mid \lambda \in \Lambda \text{ and } \xi_\lambda \in (-1, 1)^n \text{ s.t. } v' \in \mathcal{V}\}$$

is called a pseudo-periodic state with period lattice Λ

- the offset v is random
- we can prepare such pseudo-periodic states by evaluating the approximate homomorphism f

Computing the period lattice

- given access to pseudo-periodic states with unknown random offsets, the task is determine an approximate basis $\lambda'_1, \dots, \lambda'_n$ of Λ
- a basis $\lambda'_1, \dots, \lambda'_n$ is called γ -approximate iff
 - $\Lambda = \mathbb{Z}\lambda_1 + \dots + \mathbb{Z}\lambda_n$
 - $\|\lambda_i - \lambda'_i\|_2 \leq \gamma$

One-dimensional pseudo-periodic states

- let $\Lambda = R\mathbb{Z}$ be a 1d lattice with unknown period $R \in \mathbb{R}_{>0}$

- let $\mathcal{V} = \{0, \dots, qN - 1\}$

- a uniform superposition in $\mathbb{C}^{\mathcal{V}} := (\mathbb{C}^{qN})$ over a set of the form

$$\mathcal{M} = \{v' = v + mNR + \xi_m \mid m \in \mathbb{Z} \text{ and } \xi_\lambda \in (-1, 1) \text{ s.t. } v' \in \mathcal{V}\}$$

is called a pseudo-periodic state with period lattice Λ

- the offset $v \in \{0, 1, \dots, \lfloor NR \rfloor\}$ is random

- task: find a rational number $\hat{R}$ such that $|\hat{R} - R| \leq \gamma$

QA for estimating the period

- assume $q > 2R$ and $N > \frac{2}{R}$
- prepare two pseudo-periodic states in $\mathbb{C}^{\mathcal{V}}$ with period R
- embed each state into the larger space $\mathbb{C}^{\mathcal{W}}$ where $\mathcal{W} := \{0, \dots, 2qN - 1\}$
- apply the QFT to each state
- measure both states
- denote the outcomes by w_1 and w_2

Sampling approximations of Λ^*

• let $\Lambda^* = \frac{1}{R} \mathbb{Z}$ denote the dual lattice of $\Lambda = R\mathbb{Z}$

• for $\lambda^* \in \Lambda^*$, let

$$w_{\lambda^*} = [2q\lambda^*] \quad \text{rounded to closest integer}$$

• for all $\lambda^* \in \Lambda^*$ with $w_{\lambda^*} \in [0, 2q\kappa N]$ for some fixed $\kappa \in (0, 1)$, we have

$$\Pr(w_{\lambda^*}) \geq \frac{1}{2qN} \cdot \frac{q}{R} \cdot \cos^2 \left(\pi \left(\frac{1}{4} + \frac{1}{4qN} + 2\kappa \right) \right)$$

• the number of such λ^* is approximately $\frac{\kappa N}{\det(\Lambda^*)} = \kappa NR$

Approximate generating set for Λ^*

- Let a and b be two numbers chosen uniformly at random from $\{1, 2, \dots, B\}$ for $B \geq 2$

$$\Pr(\gcd(a, b) = 1) \geq 1 - \sum_{p \text{ prime}} \frac{1}{p^2} > \frac{1}{2}$$

- for half of the pairs (w_1, w_2) , the corresponding λ_i^* generate Λ^*
- $\Rightarrow$ the pair $(\frac{w_1}{2q}, \frac{w_2}{2q})$ is an approximate generating set of Λ^* with probability at least

$$\frac{\kappa^2}{8} \cos^4 \left(\pi \left(\frac{1}{4} + \frac{1}{4qN} + 2\kappa \right) \right) \geq 10^{-5} \quad \text{for } \kappa = \frac{1}{9}$$

Approximate basis for Λ

- we can recover an approximate basis of Λ^* from the approximate generating set formed by $\frac{w_1}{2q}$ and $\frac{w_1}{2q}$
- we can recover an approximate basis of Λ^* , i.e., $\hat{R}$ by taking the inverse of the approximate basis of Λ^*

Success probability of the QA

- our algorithm yields the approximation $\hat{R}$ with probability at least 10^{-5}
- Hallgren's analysis yields the significantly worse bound $\Omega(1/\log^4(M))$ where M is an upper bound on RN and works only if $\log(M) \geq 256$

$$\frac{1}{256^4} = \frac{1}{4\,294\,967\,296} < \frac{1}{4 \cdot 10^9}$$

$\Rightarrow$ the guaranteed expected running time would be at least 50 days if it took 1 ms to execute the quantum algorithm once

One-dimensional infrastructures

- the problem of computing the period lattice of a one-dimensional infrastructure



the problem of estimating the period of a one-dimensional periodic quantum state

- one-dimensional infrastructures arise from number fields of unit rank one
- quadratic number fields $\mathbb{Q}(\sqrt{d})$ with d positive and squarefree are only one type of number fields of unit rank one

Improvements

- quadratic number fields yield one-dimensional IS
⇒ our algorithms can be used to solve Pell's equation and the principal ideal problem
- in this sense, they generalize Hallgren's QA
- our improvements and extensions:
 - give a conceptually simpler presentation
 - obtain an improved technique of analyzing Fourier sampling of one-dimensional pseudo-periodic states
 - find algorithms with lower complexity and constant success probability

Pseudo-periodic states

- let Λ be a full-rank lattice in $\mathbb{R}^n$
- let $\mathcal{V} = \{0, \dots, qN - 1\}^n$
- a uniform superposition in $\mathbb{C}^{\mathcal{V}} := (\mathbb{C}^{qN})^{\otimes n}$ over a set of the form

$$\mathcal{M} = \{v' = v + N\lambda + \xi_\lambda \mid \lambda \in \Lambda \text{ and } \xi_\lambda \in (-1, 1)^n \text{ s.t. } v' \in \mathcal{V}\}$$

is called a pseudo-periodic state with period lattice Λ

- the offset v is random
- we can prepare such pseudo-periodic states by evaluating the approximate homomorphism f

Computing the period lattice

- given access to pseudo-periodic states with unknown random offsets, the task is determine an approximate basis $\lambda'_1, \dots, \lambda'_n$ of Λ
- a basis $\lambda'_1, \dots, \lambda'_n$ is called γ -approximate iff
 - $\Lambda = \mathbb{Z}\lambda_1 + \dots + \mathbb{Z}\lambda_n$
 - $\|\lambda_i - \lambda'_i\|_2 \leq \gamma$

QA for estimating the period lattice

- assume that q and N that are sufficiently large
- prepare $2n + 1$ pseudo-periodic states in $\mathbb{C}^{\mathcal{V}}$
- embed each state into the larger space $\mathbb{C}^{\mathcal{W}}$ where $\mathcal{W} = \{0, \dots, 2nqN - 1\}^n$
- apply the multidimensional QFT to each state
- measure each state
- denote the outcomes by $w_1, \dots, w_{2n+1}$

Sampling approximations of Λ^*

- let Λ^* denote the dual lattice of Λ
- for $\lambda^* = (\lambda_1^*, \dots, \lambda_n^*) \in \Lambda^*$, let

$$\mathcal{R}_{\lambda^*} = \left\{ (w_1, \dots, w_n) \mid w_k \in \{ \lfloor 2nq\lambda_k^* \rfloor, \lceil 2nq\lambda_k^* \rceil \} \right\}$$

each coordinate either rounded down or rounded up

- for all $\lambda^* \in \Lambda^*$ with $\mathcal{R}_{\lambda^*} \subset [0, 2qn\kappa N]^n$ for some fixed $\kappa \in (0, 1)$, we obtain the improved lower bound

$$\Pr(\mathcal{R}_{\lambda^*}) \geq 2^{n-1} \cdot \frac{1}{(2qnN)^n} \cdot \frac{q^n}{\det(\Lambda)} \cdot \cos^2 \left(\pi \left(\frac{1}{4} + \frac{1}{4qn} + 2\kappa \right) \right)$$

- the number of such $\lambda^* \approx \frac{(\kappa N)^n}{\det(\Lambda^*)} = (\kappa N)^n \det(\Lambda)$

Approximate generating set for Λ^*

- we sample n vectors of $\Lambda^* \cap [0, b)^n$
- with probability at least $\frac{1}{4}$ they generate a full-rank sublattice Λ_0^*
- Λ^* / Λ_0^* is a finite group generated by n elements
- we sample $n + 1$ vectors of $\Lambda^* \cap [0, b_0)^n$ with $b_0 > b$
- this induces an almost uniform distribution of Λ^* / Λ_0^*
- with probability at least $\frac{1}{4} - \hat{\zeta} > 0.184$ all the $2n + 1$ samples generate Λ^*

Approximate basis for Λ

- we can recover an approximate basis of Λ^* from the approximate generating set formed by

$$\frac{w_1}{2nq}, \dots, \frac{w_{2n+1}}{2nq}$$

- we can recover an approximate basis of Λ^* from the approximate basis of Λ^*
- the algorithm succeeds with probability at least

$$\frac{1}{4} \left(\frac{1}{4} - \hat{\zeta} \right) \left(\frac{c\kappa^n}{2n^n} \right)^{2n+1}$$

where $c = \cos^2 \left(\pi \left(\frac{1}{4} + \frac{1}{4qN} + 2\kappa \right) \right)$ and $\kappa = \frac{1}{9n}$

Improvements

- we present a rigorous and complete proof that the running time is polynomial in the determinant of the period lattice Λ
- we give an explicit lower bound on the success probability

HALLGREN and SCHMIDT AND VOLLMER did not give any lower bound

SCHMIDT presented a lower bound in his dissertation which is worse than our bound

- we improve the success probability by at least 2^{n^2-1}
- but the resulting lower bound is still quite bad $\Rightarrow$ find new ideas and techniques to improve it

Thank you!!!